

Introduction

The project is based on a survey of the Sydney Swans' members. A wide range of data including emotional attachment to the club, match day experience, membership renewal, and many more was collected. This provides invaluable information to understand Swans members' attitudes, motivations, and drivers of fan loyalty.

Our core objective was to **improve the Sydney Swans' business performance**, with a focus on increasing membership renewal rates. Membership is the club's primary source of revenue which means increasing renewal rates directly improves financial performance and long-term sustainability. In this context, our key **research question** was: **What could influence the renewal of Sydney Swans membership?** It is vital to understand this link, as fan engagement and loyalty are not only emotional assets, but also financial drivers for the club.

The survey data offers potential to explore additional questions:

- What emotional, attitudinal and experiential factors best predict member renewal?
- How do members rate the Swan's performance in terms of atmosphere, communication, facilities and engagement?
- What changes or improvements would make members more likely to renew?
- Do demographic or behavioral characteristics (e.g., age, attendance history) affect the likelihood of renewal?

While the data can address all of the above questions, only the main key research question regarding membership renewal will be our focus.

Data Preprocessing and Features Selection

Our target variable (Y) is q3: "Has your level of support for the Swans changed since becoming a Swans member?". This question is a key indicator of fan loyalty and is

closely linked to their likelihood of renewing membership. Responses are recorded on a Likert scale ranging from 1 (much weaker) to 7 (much stronger), with 4 indicating no change. To simplify prediction, Likert scales were transformed into a binary classification. Responses with a score above 4 (moderately to strongly support) were labeled as 'Likely to Renew', while those with 4 or below were labeled 'Unlikely to Renew'. Missing values in q3 were imputed with 1 under the assumption that the remaining responses cover the range from 2 to 7.

For data types, most of our features are either numerical (e.g., age, number of games attended) or ordinal categorical (e.g., Likert scale responses). Some are binary flags (e.g., whether a person attended a particular game), and a few are percentage distributions (like q81 to q85).

Missing values were addressed. For age-related features (q12_age_int, q13_age_attd), if one was missing, the other's value was used for imputation. If both were missing, the median age of the corresponding age group was used. Missing values in q14_no_game were replaced with median due to right skewness. Categorical features (q15_team_sup, q16_attd_who) were imputed with '4' as a category representing 'unknown' or 'other'. For the remaining features, either modes or a category representing unknown were used for imputation. For question groups: q2, q6, q7, q8, q9, missing subcategory values were imputed using base category's value. For example, if q21 is missing, we impute with value from q2. For question groups: q6, q8, where the respondents assigned 100 points across sub-questions, totals exceeding 100 were resolved by either dropping rows (for q8) or retaining them (for q6, as dropping worsened the model). We also observed that survey responses followed ordinal or binary formats, with clear peaks at extreme values (e.g., 1 or 7 on Likert scales), indicating strong opinions from respondents.

For feature selection, we examined relationships between all variables and the binary target variable Y. Correlations of all variables with the target variables were considered. Several features, such as emotional connection variables (e.g., q52, q58, and q144)

showed moderate correlation with Y, suggesting that fan loyalty and pride are predictive of renewal behavior.

We also used logical reasoning to guide our choices—for example, it's reasonable to expect that the q8 questions (reasons for supporting the team) are associated with fan loyalty (q3).

Additionally, iterative modeling experiments were conducted. We tested modelling with different feature combinations to assess predictive performances.

By combining insights from EDA results, common sense, and these modeling experiments, we selected features that not only performed well but also made practical sense in the context of membership renewal.

Final selected features include variables from q8, q9, q5, q12a, q13, and q15 (detailed in Appendix). These questions capture emotional connection, behavioral patterns, and attitudinal responses, which we believe are strong indicators of fan loyalty and likelihood to renew membership. Many of them are based on Likert scales, allowing us to measure the strength of attitudes numerically.

Modeling

To predict the likelihood of Sydney Swans membership renewal, we employed five classification models: Random Forests, Logistic Regression, Support Vector Machines (SVM), XGBoost, and Bagging.

Our initial approach involved a standard dataset split of 80% training and 20% testing. However, this resulted in poor model performance of accuracy at 52% - 64%. (Table 3, Appendix). Further investigation revealed that was due to a significant class imbalance in the target variable: 68.84% were classified as 'likely to renew', leaving only 31.16% in the 'unlikely to renew' category (Table 1). This imbalance hindered models' ability to accurately predict the minority class (Buda, Maki and Mazurowski, 2018).

Table 1: Distribution of Target Variable

Category	Count	Percentage
Likely to Renew	464	68.84%
Unlikely to Renew	210	31.16%
Total	674	100%

To address this, the Synthetic Minority Over-sampling Technique (SMOTE) (Chawla et al., 2002) was employed. By generating synthetic samples for the 'unlikely to renew' category, SMOTE created a more balanced training data, leading to better model performance. Consequently, our subsequent analysis will focus on models using SMOTE.

Bagging, tested with decision trees and random forests as base estimators, performed better using random forests (Table 4, Appendix).

To optimize the Random Forest model, hyperparameter tuning was performed using BayesSearchCV. Table 5 (Appendix) shows the hyperparameters and their optimal values, which improved the model's accuracy from 81.52% to 82.61%:

For Logistic Regression and SVM, feature standardization was explored. However, it did not enhance models' performance.

Interpretation of Results and Feature Importance

Table 2: Model's performance summary

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC score
Random Forest	0.8261	0.8200	0.8542	0.8367	0.8248
Logistic Regression	0.6685	0.6923	0.6562	0.6738	0.6690
SVM	0.6739	0.6837	0.6979	0.6907	0.6728
XGboost	0.8098	0.8020	0.8438	0.8223	0.8082
Bagging	0.8424	0.8384	0.8646	0.8513	0.8414

Shown in Table 2, Bagging achieved the best performance (accuracy: 84.24%, F1-score: 0.8513, ROC-AUC: 0.8414), demonstrating effectiveness at balancing precision and recall, and class distinction. Random Forest and XGBoost also showed strong predictive capabilities, with F1-scores of 0.8367 and 0.8223, and ROC-AUC scores of 0.8248 and 0.8082.

However, Logistic Regression and SVM underperformed. Logistic Regression achieved 66.85% accuracy, while SVM achieved 67.39%. Their F1-scores of 0.6738 and 0.6907 indicate moderate predictive capability.

Feature importance for top three best-performing models—Bagging, Random Forest, and XGBoost—were further examined.

Shown in Figure 1, 2, 3 and Table 6, 7, 8 (Appendix), Bagging identified top three predictors as: q85 (providing good facilities for watching the game), q933 (organising functions and social opportunities), and q921 (creating an exciting atmosphere).

Random Forest highlighted q921, q85, and q941 (professional marketing and management of the club) as the top three most crucial features.

Meanwhile, XGBoost showed that q941, q92 (delivering a better overall game experience), and q921 were the top three key drivers.

Across models, q921 was consistently in the top-three most important predictors. Additionally, q85 was a key predictor in both Bagging and Random Forest. This underscores the importance of an exciting atmosphere and good facilities.

Figure 1: Top 10 features with highest importance scores (Bagging)

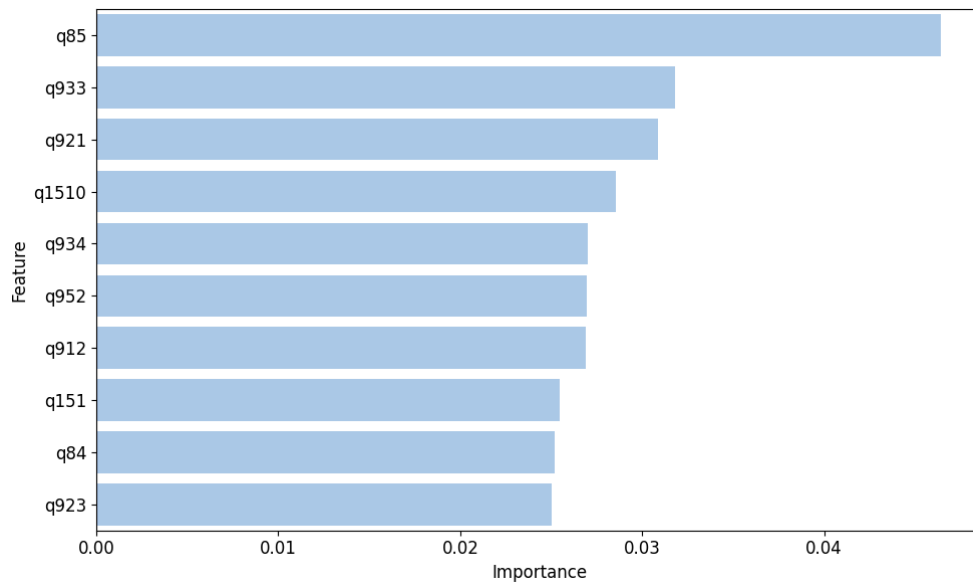


Figure 2: Top 10 features with highest importance scores (Random Forest)

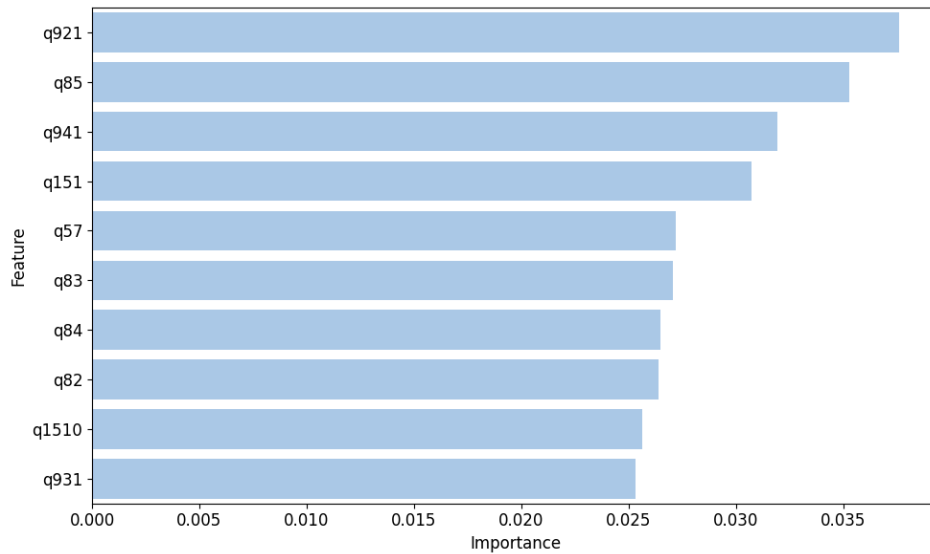
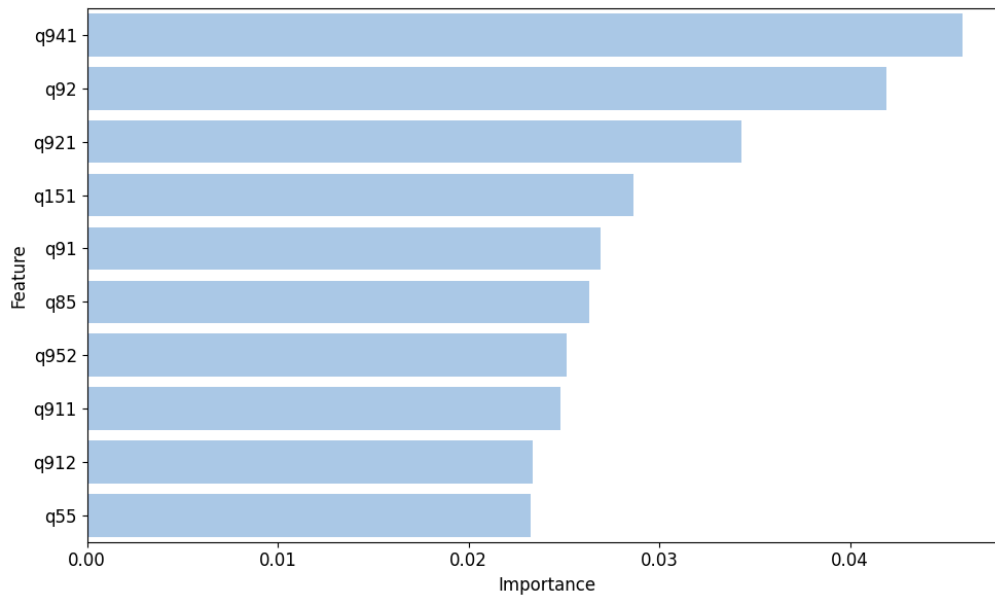


Figure 3: Top 10 features with highest importance scores (XGBoost)



Recommendation

According to modelling results, Bagging performed the best. We propose the following strategies based on the bagging model's top feature importance to improve Sydney Swans' membership renewal rates and overall business performance:

- Focus on improving match day vibe

Q921 (Creating an exciting atmosphere) consistently ranked in the top three of all high performance models, suggesting that the stadium atmosphere plays a critical role in fans' emotional attachment and renewal decisions. We recommend that the Swans invest in immersive entertainment, spectator interactions and fan rituals to create an engaging environment.

- Improve spectator facilities

Q85 (Provide good facilities from which to watch the game) is also a top predictor in both Bagging and Random Forest models, indicating importance of facilities. Thus, we suggest improving stadium seating comfort, food and beverage options, visual accessibility, and cleanliness. This is expected to significantly improve overall fan satisfaction and increase the likelihood of repeat visits. As Shilbury et al. (2014) highlight in sport marketing research, managing the service environment is one of the key factors in developing high levels of customer satisfaction.

- Increased socialization and community involvement

Q933 (Organize functions and social opportunities) being a top feature of bagging highlights the importance of spaces for fans to connect outside of game day. Members-only events, fan forums, and opportunities to meet players or staff can enhance a sense of community and deepen loyalty.

Limitations

While the project produced meaningful insights and robust model performance, limitations remain:

- Synthetic samples

SMOTE helped balance the data and improved performance. However, synthetic samples creation may introduce artificial patterns that do not exist in the real-world population, which may affect generalizability.

- Binary simplification of membership intention (Q3)

Target variable Q3 was originally measured using a 7-point Likert scale, but for modeling purposes we simplified it to a binary variable. This approach makes prediction feasible, but may mask nuances in fan sentiment, especially among neutral respondents who scored a “4”.

- Based solely on survey data

The analysis relies entirely on self-reported survey responses. It lacks behavioral data, such as actual attendance records, purchase history, or engagement on digital platforms, which would provide a more complete picture of update behavior.

A broader range of data should be collected for better analysis of member behavior.

References

Buda, M., Maki, A. and Mazurowski, M. A. (2018). A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, 106, pp. 249-259. doi:<https://doi.org/10.1016/j.neunet.2018.07.011>

Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16(16), pp.321–357. doi:<https://doi.org/10.1613/jair.953>.

Shilbury, D., Westerbeek, H., Quick, S., Funk, D. & Karg, A. (2014). Customer Satisfaction and Service Quality. In: *Sport Marketing*. Publisher: Routledge. Pp.21. URL: <https://www.taylorfrancis.com/chapters/mono/10.4324/9781003117483-11/customer-satisfaction-service-quality-david-shilbury-hans-westerbeek-shayne-quick-daniel-funk-adam-karg?context=ubx>

Appendix

Tables

Table 3: Model Performance of using Normal Splitting approach

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC score
Random Forest	0.6316	0.6270	0.9753	0.7633	0.5357

Logistic Regression	0.6466	0.6417	0.9506	0.7662	0.5618
SVM	0.5263	0.6286	0.5432	0.5828	0.5216
XGboost	0.6241	0.6220	0.9753	0.7596	0.5261
Bagging	0.6316	0.6250	0.9877	0.7656	0.5323

Table 4: Bagging Performance by Base Estimator (using SMOTE)

Base estimator	Accuracy	Precision	Recall	F1-score	ROC-AUC score
Decision Tree	0.8207	0.8247	0.8333	0.8290	0.8201
Random Forest	0.8424	0.8384	0.8646	0.8513	0.8414

Table 5: Hyperparameter tuning of Random Forest

Hyperparameter	Optimal Value
bootstrap	FALSE
class_weight	balanced
criterion	entropy
max_depth	100
max_features	sqrt
min_impurity_decrease	0
min_samples_leaf	1
min_samples_split	2
n_estimators	200

Table 6: Top 10 Features - Bagging (SMOTE with base estimator as Random Forest)

Feature	Importance
q85	0.0464
q933	0.0318
q921	0.0309
q1510	0.0286
q934	0.027
q952	0.027
q912	0.0269
q151	0.0255
q84	0.0252
q923	0.025

Table 7: Top 10 Features - Random Forest (SMOTE with tuned hyperparameters)

Feature	Importance
q921	0.0376
q85	0.0352
q941	0.0319
q151	0.0307
q57	0.0272
q83	0.0271
q84	0.0265
q82	0.0264
q1510	0.0257
q931	0.0253

Table 8: Top 10 Features - XGBoost (SMOTE)

Feature	Importance
q941	0.0459
q92	0.0419
q921	0.0343
q151	0.0286
q91	0.0269
q85	0.0263
q952	0.0252
q911	0.0248
q912	0.0234
q55	0.0233

Detailed questions of Selected Features

Q5 Questions

“Please indicate the extent to which each of the following describes you. 7 means it describes you very well, 6 if it describes you quite well and so on to 2 if it describes you a little and 1 if it describes you not at all.”

- **q51:** It's important to me to maintain ties with the Sydney Swans.
- **q52:** Supporting the Swans is an important part of my life.
- **q53:** I tell others that I am a fan of the Swans.
- **q54:** I tell others that I am a member of the Sydney Swans.

- **q55:** I think of myself as part of the Sydney Swans organization.
- **q56:** I feel I have more in common with other members than non-members.
- **q57:** When the Swans are doing well, I feel that it is partly my accomplishment.
- **q58:** My whole week is better if the Swans have won their last game.

Q8 Questions

"How important is each of the following to you in assessing the Swans organization?"

Respondents were asked to allocate 100 points among following statements:

- **q81:** Try harder for members (e.g., Try to give me benefits that non-members don't get, including access to finals tickets).
- **q82:** Are able to provide a better game experience (e.g., Create an exciting atmosphere./ Provide me with good seats in the stadium /Provide quality and variety of entertainment/Promote a family atmosphere).
- **q83:** Increase my accessibility to the club (e.g., Make me feel like part of the club/Keep me up to date on club matters./Organise functions and opportunities and the chance to mingle with players).
- **q84:** Efficiently manage the club (e.g., Professionally market and manage the Club./Employ courteous and professional staff/Maintain a positive image of the club).
- **q85:** Provide good facilities from which to watch the game. (e.g., A comfortable stadium/ A visually appealing stadium./ A clear and accurate scoreboard).

Q9 Questions

"Now, please indicate how well you believe the Swans perform with regard to the following general/overall and specific issues, using a 7-point scale where 7 = Perform Very Well, 4 = perform neither well nor poorly and 1 = Perform Very Poorly."

- **q91:** Overall, Try Harder for Members
 - q911 - Try to give members benefits that non-members don't get.
 - q912 - Try to get members finals tickets.

- **q92:** Overall, Are able to provide a better game experience
 - q921 - Create an exciting atmosphere.
 - q922 - Provide members with good seats in the stadium.
 - q923 - Provide quality and variety of entertainment at the games.
 - q924 - Promote a family atmosphere at the games.
- **q93:** Overall, Increase my accessibility to the club
 - q931 - Make me feel like part of the club.
 - q932 - Keep me up to date on club matters.
 - q933 - Organise functions and social opportunities.
 - q934 - Give me the chance to mingle with players).
- **q94:** Overall, Efficiently manage the club
 - q941 - Professionally market and manage the club.
 - q942 - Maintain a positive image of the club.
 - q943 - Employ courteous and professional staff).
- **q95:** Overall, Provide good facilities with which to watch the game.
 - q951 - A visually appealing stadium.
 - q952 - A comfortable stadium.
 - q953 - A clear and accurate scoreboard.

Q12a Question

- **q12a:** Indifference
 - This is the subquestion from Q12, which is "And finally, consider a few negative feelings and whether you have ever had these about the Swans."

Q13 Questions

"Please indicate the extent to which you agree or disagree with each of the following statements (7 indicates that you strongly agree, 6 indicates that you agree but not quite so strongly, 4 indicates that you neither agree or disagree, 1 indicates that you strongly disagree)."

- **q131:** I have confidence in the accuracy of the information I get from the Swans.
- **q132:** I have confidence in the fairness and honesty of the Swans management.
- **q133:** I can always rely on being informed early enough by Swans management about new products or issues I might be interested in.
- **q134:** Swans management behaves in a trustworthy manner towards me.
- **q135:** Swans management is only concerned with promoting their own interests.
- **q136:** Swans management cares about Swans members.
- **q137:** Swans management has the capability to run an effective Supporter organization.
- **q138:** I trust Swans management.
- **q139:** Swans management are trying hard to run an effective Supporter organization.

Q15 Questions

"This question asks you to indicate how a number of hypothetical situations might influence your level of support for the Swans. Please indicate the extent of your likelihood, in regard to each of the following statements. Circle 7 if you are very likely, 4 if you are neither likely nor unlikely, and 1 if you are very unlikely."

- **q151:** I would be more likely to continue to be a member if facilities at games improved.
- **q152:** I would continue to be a member if membership fees rose by 30%.
- **q153:** I would continue to be a member if the Swans came last on the ladder for one or more seasons.
- **q154:** I would continue to be a member if the Swans moved to Homebush and played all home games there.
- **q155:** I would continue to be a member if membership fees rose by 10%.
- **q156:** I would continue to be a member if I got worse seats than I have now.
- **q157:** I would continue to be a member if seats were readily available for all home games without being a member.
- **q158:** I would continue to be a member if the Swans moved to another city.

- **q159:** I would continue to be a member if the Swans played half of their home games at Homebush.
- **q1510:** I would be more likely to continue to be a member if I had more say in the way the members' organization was run.